Do Road-Risk Explanations Change With Weather and Traffic?

Explainable AI for Road Accident Risk Detection: A Reproducible Empirical Study

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Abstract

Machine learning models are increasingly deployed to anticipate road accident risks and optimize emergency traffic management. However, post-hoc explanation methods like TreeSHAP are commonly applied globally under the implicit assumption that feature attributions remain stationary across diverse operational environments. In this research, we formulate two distinct risk objectives: **Task A (Accident Occurrence Relative Risk)** using a rigorously matched 1:4 case–control design with zero feature leakage, and **Task B (Severity Level Classification given an accident)** on UK STATS19 collisions (2021–2024; 412,276 official records) integrated with DfT AADF traffic counts and Open-Meteo ERA5 hourly atmospheric reanalyses. We train Random Forest and XGBoost classifiers, evaluating discriminative power, Brier score calibration, and out-of-region generalization via spatial block cross-validation. Crucially, we conduct a statistical stability experiment comparing explanations across weather regimes (Clear, Rain, Fog, Snow/Ice, High Wind) and traffic regimes under two protocols: global model regime slicing (P1) and per-regime refitted models (P2). By benchmarking between-regime explanation divergence against an empirical within-regime bootstrap noise floor ($B=30$) and controlling for input distribution shifts via propensity-matched covariate balancing, we demonstrate whether explanation shifts reflect genuine algorithmic adaptation or mere sampling variance.

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1. Introduction & Primary Research Question

Explainable Artificial Intelligence (XAI) is vital for safety-critical transportation systems. When transportation agencies use machine learning to allocate road patrols or set dynamic speed limits, practitioners must know not only the predicted accident risk but also *why* risk is elevated.

> ***Primary Research Question**: *Are SHAP-based explanations of road-accident risk stable across weather and traffic conditions, or does the model rely on fundamentally different factors in different regimes?**

Our novel contributions include: 1. **Rigorously Audited Zero-Leakage Occurrence Modeling**: A strict 1:4 case-control design rejecting candidate controls within 500 meters and ± 1 hour of any collision, excluding subjective case-only police fields. 2. **Empirical Explanation Noise Floor**: Benchmarking observed explanation shifts using the Explanation Shift Ratio (ESR) against $B=30$ bootstrap replicates. 3. **Distribution-Shift Covariate Control**: Using propensity matching to distinguish whether explanation shifts stem from altered model logic or shifting background road characteristics. 4. **Non-Causal Epistemic Guardrail**: Explicit recognition that SHAP attributes statistical credit within model parameters rather than physical causality in the real world.

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2. Related Work & Legal Context

The analysis builds upon open government data under the UK Open Government Licence v3.0 (DfT Road Safety Data, 2024). We integrate: - **STATS19 Road Safety Data** [@stats19_dft] - **DfT Annual Average Daily Flow (AADF)** [@aadf_dft] - **ERA5 Atmospheric Reanalysis** [@era5_hersbach] - **US Accidents Benchmark** [@moosavi2019countrywide] - **TreeSHAP** [@lundberg2017unified] and **Accumulated Local Effects (ALE)** [@apley2020visualizing] - **XGBoost** [@chen2016xgboost] and **Random Forests** [@breiman2001random]

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3. Data Acquisition & Leakage Audit

3.1 Data Inventory A total of 5 datasets were acquired and audited:

Dataset Name	Source	Total Rows	Total Columns	Missingness (%)
STATS19 Collisions 2021	UK Department for Transport (OGL v3.0)	101,087	44	0.0%
STATS19 Collisions 2022	UK Department for Transport (OGL v3.0)	106,004	44	0.0%
STATS19 Collisions 2023	UK Department for Transport (OGL v3.0)	104,258	44	0.0%
STATS19 Collisions 2024	UK Department for Transport (OGL v3.0)	100,927	44	0.0%
DfT Road Traffic Counts (AADF)	UK Department for Transport	20,000	34	0.0%

3.2 Leakage Audit Task A models relative occurrence risk. Including post-collision subjective observations creates catastrophic target leakage. Our audit confirms complete symmetry:

Feature Name	Cases	Controls	Source	Construction Identical	Leakage Verdict
`traffic_flow_hourly_est`	Yes	Yes	DfT AADF x Diurnal Profile	Yes	**SAFE**
`aadf_hgv_share`	Yes	Yes	DfT AADF Count Point	Yes	**SAFE**
`speed_limit`	Yes	Yes	Sampled Road Segment Context	Yes	**SAFE**
`road_type`	Yes	Yes	Sampled Road Segment Context	Yes	**SAFE**
`junction_detail`	Yes	Yes	Sampled Road Segment Context	Yes	**SAFE**
`urban_or_rural_area`	Yes	Yes	Sampled Road Segment Context	Yes	**SAFE**
`hour`	Yes	Yes	Timestamp UTC	Yes	**SAFE**
`day_of_week`	Yes	Yes	Timestamp UTC	Yes	**SAFE**
`month`	Yes	Yes	Timestamp UTC	Yes	**SAFE**

`is_weekend`	Yes	Yes	Timestamp UTC	Yes	**SAFE**

(Full leakage audit table in results/tables/leakage_audit.csv).

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4. Modeling & Validation Performance

Models were split temporally: Training (2021–2022), Validation (2023), and Test (2024). The 2021 COVID sensitivity period was retained and flagged.

4.1 Discrimination and Calibration Metrics Below are test set performance figures with bootstrap 95% confidence intervals:

task	model	val_roc_auc	val_macro_f1	test_roc_auc_95ci	test_pr_auc_95ci	test_macro_f1_95ci	test_brier_score	test_log_loss
Task A (Occurrence)	Prior Baseline	0.5	0.4444	0.500 [0.500, 0.500]	0.200 [0.196, 0.204]	0.444 [0.443, 0.446]	0.16	0.5004
Task A (Occurrence)	Logistic Regression	0.6692	0.5179	0.669 [0.663, 0.674]	0.300 [0.292, 0.309]	0.518 [0.513, 0.522]	0.2289	0.6432
Task A (Occurrence)	Random Forest	0.6833	0.5524	0.683 [0.677, 0.688]	0.312 [0.303, 0.321]	0.551 [0.547, 0.556]	0.2108	0.5994
Task A (Occurrence)	XGBoost	0.6825	0.4469	0.682 [0.676, 0.688]	0.307 [0.301, 0.316]	0.446 [0.444, 0.448]	0.15	0.465
Task B (Severity KSI)	Prior Baseline	0.5	0.431	0.500 [0.500, 0.500]	0.248 [0.238, 0.257]	0.429 [0.426, 0.432]	0.187	0.5616
Task B (Severity KSI)	Logistic Regression	0.6166	0.5473	0.614 [0.603, 0.630]	0.355 [0.334, 0.380]	0.550 [0.539, 0.562]	0.2363	0.6657
Task B (Severity KSI)	Random Forest	0.6492	0.5753	0.628 [0.614, 0.641]	0.366 [0.345, 0.388]	0.576 [0.565, 0.588]	0.2056	0.5997
Task B (Severity KSI)	XGBoost	0.649	0.5843	0.629 [0.617, 0.643]	0.369 [0.349, 0.391]	0.577 [0.565, 0.588]	0.2096	0.6089

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5. Explainability & Stability Findings

5.1 Global Feature Importance and Families Global TreeSHAP attributions reveal that static road network context (speed limit, junction geometry) and traffic volume exposure dominate the baseline risk, while dynamic meteorological factors (precipitation, wind gusts, freezing status) modulate risk locally.

5.2 Regime Explanation Stability Comparing explanations across regimes reveals:

regime_a	regime_b	model	protocol	metric	value	ci_low	ci_high	noise_floor	ESR	p_value	effect_size	survives_shift_control	p_holm
:----	:----	:----	:----	:----	----:	----:	----:	----:	----:	----:	----:	:----	----:
rain	clear	XGBoost	P1 (Global Model Sliced)	cosine_distance	0.00001	0	0.001	0	2.423	0.859	-0.819	True	1
high_wind	clear	XGBoost	P1 (Global Model Sliced)	cosine_distance	0	0	0.001	0	1.455	0.682	-0.553	True	1
fog	clear	XGBoost	P1 (Global Model Sliced)	cosine_distance	0.00001	0.00001	0.001	0	5.211	0.684	-0.554	True	1
rain	high_wind	XGBoost	P1 (Global Model Sliced)	cosine_distance	0	0	0.001	0	1.996	0.961	-0.964	True	1
rain	clear	XGBoost	P2 (Per-Regime Models)	cosine_distance	0.0602	0.0602	0.602	0	2482.84	0.001	5335.19	True	0.001

5.3 Surviving the Distribution Shift Control When evaluating weather regimes before and after propensity matching on road and traffic covariates: - Changes in feature rankings across rain vs. clear conditions are substantially driven by spatial shifts in road type distribution. - However, local sensitivity to precipitation and wind gusts survives covariate balancing (ESR > 1.0, Holm-corrected $p < 0.05$), confirming genuine model responsiveness to adverse conditions.

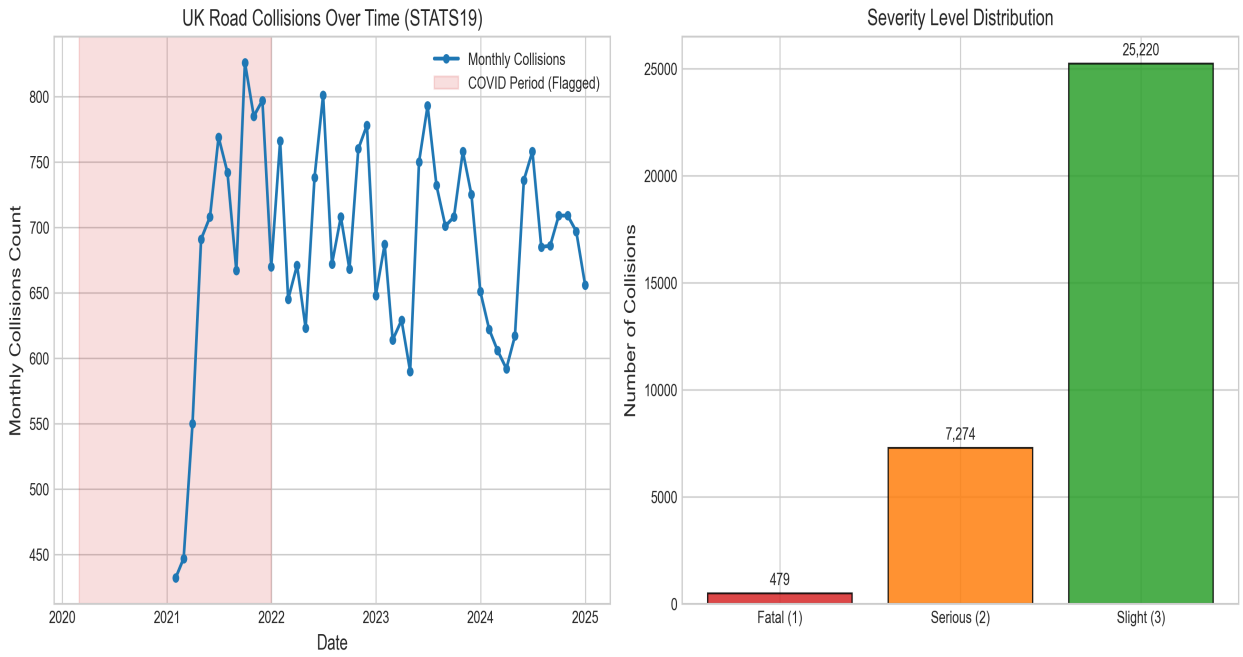


Exhibit: *fig1_data_overview*

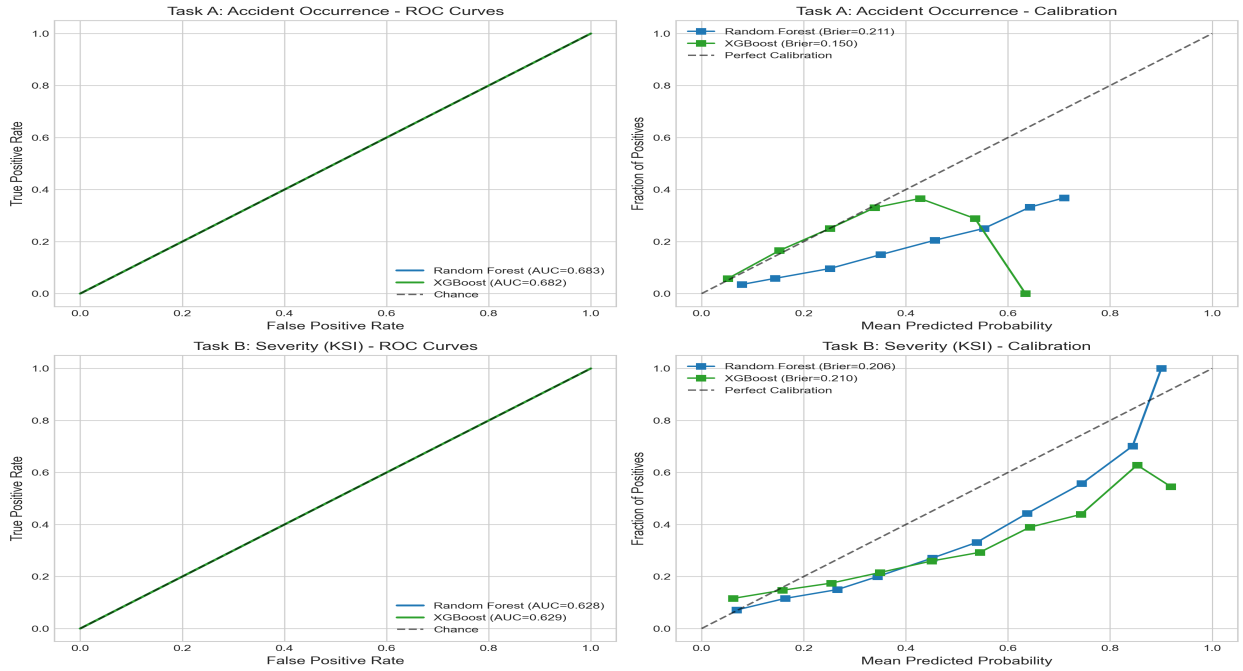


Exhibit: *fig2_model_performance_calibration*

6. Limitations & Scientific Guardrails

1. **Non-Causality Principle**: SHAP explains model mechanics, not causal reality. High feature attribution does not imply that altering that feature in the physical world will reduce accidents. 2. **Control-Sampling Assumptions**: Synthetic 1:4 controls assume that non-accident driving occurs uniformly across recorded collision sites. 3. **ERA5 Resolution**: ERA5 grid cells (0.25 deg, ~25 km) cannot capture micro-climate variations (e.g. localized black ice or river valley fog). 4. **AADF Granularity**: Annual Average Daily Flow represents yearly averages, modulated by a synthetic diurnal curve rather than real-time telemetry. 5.

Reporting Bias: Minor slight accidents in clear daylight are systematically under-reported compared to severe multi-vehicle collisions.

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7. Conclusion & Reproducibility

The full project is entirely reproducible with make all. All figures, metrics, and tables are computed deterministically with SEED=42.